

Senyi Kong

PhD student in Earth Climate and Environment

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ResearchGate | Expected PhD completion: June 2027

RESEARCH PROFILE

Atmospheric scientist studying electromagnetic scattering by nonspherical and inhomogeneous particles, aerosol and cloud optics, hyperspectral satellite retrievals, and data assimilation for numerical weather prediction. Research spans physics-based algorithm development, satellite applications, and forecast.

EDUCATION

PhD in Earth Climate and Environment | Zhejiang University

Sep 2022 - Jun 2027 (expected) | School of Earth Sciences.

BSc in Atmospheric Science | Zhejiang University

Sep 2018 - Jun 2022 | GPA 4.60/5.00; ranked 1st of 15.

RESEARCH PROJECT CONTRIBUTIONS

All-sky assimilation of FY-4 hyperspectral observations

Jan 2025 - Dec 2028 | National Natural Science Foundation of China (NSFC), Joint Fund

Participant, Investigated assimilation methods for FY-4 infrared hyperspectral satellite observations.

Electromagnetic scattering by large nonspherical particles

Jul 2023 - Jun 2026 | Chinese Academy of Sciences, major research infrastructure open-access project

Participant, Computed electromagnetic scattering by large atmospheric particles.

All-weather Fengyun satellite data assimilation

Nov 2022 - Oct 2025 | National Key Research and Development Program of China

Participant, Contributed to all-sky assimilation of geostationary hyperspectral infrared data within a global assimilation framework.

Fast scattering calculations for cloud-affected FY-4 observations

Aug 2022 - May 2025 | CMA Earth System Modeling and Prediction Centre, collaborative project

Participant, Developed fast scattering models for nonspherical cloud particles in infrared hyperspectral applications.

FY-4 data assimilation for high-impact weather forecasting

Jul 2023 - Dec 2023 | Zhejiang Institute of Meteorological Sciences, collaborative project

Participant, Worked on atmospheric profile retrieval and data assimilation for CMA-MESO.

Scattering by nonspherical, inhomogeneous atmospheric particles

Jan 2021 - Dec 2023 | NSFC, Excellent Young Scientists Fund project

Participant, Modelled dust particles and calculated their optical properties.

PUBLICATIONS

1. X. Song, W. Han, **S. Kong**, J. Yan, K. Zhou (2026). Reconstruction of Three-Dimensional Cloud Structures Using FY-4B AGRI Observations and CMA-MESO Forecasts. *Weather and Forecasting*, Early online release, 14 August 2026.
10.1175/WAF-D-25-0251.1
2. L. Bi, **S. Kong**, Y. Xi, X. Wang, W. Lin, Z. Du, W. Han, X. Zhang (2026). ZJU-SCATT-V1.0: A Comprehensive Single-Scattering Database of Nonspherical Aerosol and Cloud Particles. *Bulletin of the American Meteorological Society*, 107(7), E1563-E1578.
10.1175/BAMS-D-25-0258.1
3. **S. Kong**, W. Han, L. Bi, R. Yin (2026). All-sky FY4B-GIIRS infrared hyperspectral satellite data assimilation in the CMA-GFS 4DVar system. Part I: Implementation. *Quarterly Journal of the Royal Meteorological Society*, 152(778), e70185.
10.1002/qj.70185

4. **S. Kong**, W. Han, L. Bi, R. Yin (2026). All-sky FY4B-GIIRS infrared hyperspectral satellite data assimilation in the CMA-GFS 4DVar system. Part II: Evaluation. *Quarterly Journal of the Royal Meteorological Society*, 152(778), e70184.
10.1002/qj.70184
5. **S. Kong**, L. Bi, W. Han, R. Yin, H. Zhang (2026). Hybrid Bayesian-Machine Learning Framework for Multi-Profile Atmospheric Retrieval from Hyperspectral Infrared Observations. *Advances in Atmospheric Sciences*, 43(2), 373-389; online November 2025.
10.1007/s00376-025-5070-9
6. Zhang, H., Han, W., Luo, L., **Kong, S.**, Chen, F., & Shen, H. (2026). Thermodynamic Diagnosis and Humidity Assimilation Using FY-4B GIIRS Retrieved Temperature and Humidity Profiles for Early Warning and Forecasting of a Summer Thermal Convection Event. *Journal of Meteorological Research*. <https://doi.org/10.1007/s13351-026-5287-2>
7. R. Yin, W. Han, Z. Zhao, X. Gong, **S. Kong**, F. Zhang, J. Li (2025). The Preliminary Assimilation and Impacts of FY-4B Geostationary Interferometric Infrared Sounder (GIIRS) Cloud-Cleared Radiances. *Journal of Geophysical Research: Atmospheres*, 130(23), e2025JD043520.
10.1029/2025JD043520
8. **S. Kong**, Z. Wang, L. Bi (2024). Uncertainties in laboratory-measured shortwave refractive indices of mineral dust aerosols and derived optical properties: a theoretical assessment. *Atmospheric Chemistry and Physics*, 24, 6911-6935.
10.5194/acp-24-6911-2024
9. Z. Wang, L. Bi, **S. Kong** (2023). Flexible implementation of the particle shape and internal inhomogeneity in the invariant imbedding T-matrix method. *Optics Express*, 31(18), 29427-29439.
10.1364/OE.498190
10. **S. Kong**, K. Sato, L. Bi (2022). Lidar Ratio-Depolarization Ratio Relations of Atmospheric Dust Aerosols: The Super-Spheroid Model and High Spectral Resolution Lidar Observations. *Journal of Geophysical Research: Atmospheres*, 127(4), e2021JD035629.
10.1029/2021JD035629

PATENT

An iterative bias-correction method accounting for non-uniformity among satellite instrument detector elements.

Granted invention patent. Second-listed inventor. Application CN202511848925.6; filed December 2025.

HONOURS AND AWARDS

2025 | Labour Education Role Model, Zhejiang University

2024; 2023 | Outstanding Graduate Student, Zhejiang University

2023 | Graduate Student Award for All-Round Excellence, Zhejiang University

2023 | Top Ten Doctoral Research Advances, School of Earth Sciences, Zhejiang University

2022 | Top Ten Poster Award, School of Earth Sciences, Zhejiang University

2018-2022 | National Scholarship; First-Class Scholarship, Zhejiang University

2018-2022 | Outstanding Graduate, Zhejiang Province; Outstanding Youth League Cadre and Member, Zhejiang University

2019-2021 | First Prize, 11th National Undergraduate Mathematics Competition (non-mathematics category); Second Prize, Zhejiang Undergraduate Physics Innovation Competition (theory)